

Process Walkthrough

1. Dataset Familiarization

a. Dataset Structure & Variables

- The dataset consists of **3,847 rows** and **6 columns**.
- The columns are:
 - PRODUCTID: Unique identifier for each product.
 - TITLE: The full name/title of the product.
 - BULLET_POINTS: Features/highlights of the product.
 - DESCRIPTION: Detailed text about the product.
 - PRODUCTTYPEID: A numeric category ID.
 - ProductLength: A physical measurement of the item.
- Only the ProductLength column is numerical/quantitative. The rest are categorical

b. Significant Variables for Cleaning

- TITLE: This is the primary variable for creating the short_title. It contains a lot of "noise" (sizes, brand names, redundant descriptions) that we'll need to strip away.
- BULLET_POINTS, DESCRIPTION, PRODUCTTYPEID, and ProductLength have missing values
- All columns in the dataset were stored in 'General' format in Excel.

c. Column Consistency

- Casing Issue: All column names are in uppercase except ProductLength.
- BULLET_POINTS is the only column name with an underscore
- Missing Values: BULLET_POINTS: 1591, DESCRIPTION: 2144, PRODUCTTYPEID:178, ProductLength: 178
 - I used the filter functionality to ascertain if a column has missing values, and used the COUNTBLANK() function to get the count.

2. Initial Data Exploration

a. Descriptive Statistics

- Total Records: 3847
- Unique Products: 3,541 (This indicates we have repeat entries for the same ID).
 - The duplicates were counted using the PRODUCTID column, because it is the unique identifier. I created a column (away from the dataset) called Duplicate_Check, then I applied this formula to it:

=IF(COUNTIF(\$A\$2:A2, A2) > 1, "Extra Copy", "Unique")

After that, I aggregated the count of duplicates using

=COUNTIF(H2:H3848, "Extra Copy")

I arrived at a count of 306. This means there are 306 repeated entries in the PRODUCTID column.

- For ProductLength (The quantitative column), the relevant summary statistics were obtained using Excel's "Analysis ToolPak" feature.
 - Mean: 1150.53
 - Median: 640
 - Mode: 600
 - Standard Deviation: 2665.90
 - Maximum: 96000 (A significant outlier compared to the median).
 - Minimum: 1
 - Count: 3669

b. Data Quality Issues

- i. **Missing values:** DESCRIPTION is missing in 55% of rows (2,144 nulls). BULLET_POINTS is missing in 41% (1,591 nulls). ProductLength and PRODUCTTYPEID each have 178 (4%) missing values.
- ii. **Duplicates:** There are 306 instances where a PRODUCTID (the unique identifier) appears more than once, even if other columns slightly differ.
- iii. **Inconsistent formats:** There are slight inconsistencies in the naming format of the columns. Also in the Excel sheet, all columns were stored as "General" data type.

3. Data Cleaning Process

a. Handling Missing Values

- For the BULLET_POINTS and DESCRIPTION columns (which had high null counts), I replaced missing values with "Data Not Available" placeholders. This preserves the rows for analysis while making it clear that text data is missing. I used IF() function to achieve this
- I noticed that in rows where PRODUCTTYPEID is null, ProductLength, BULLET_POINTS, and DESCRIPTION are also null. Only corresponding PRODUCTID and TITLE are populated.
- As such, For PRODUCTTYPEID and ProductLength, I identified that rows missing these values are generally unusable for category-based analysis. I removed these sparse rows (around 4%) to maintain high data quality.

b. Removing Duplicate Entries

- ID Duplicates: After removing the rows with nulls in `PRODUCTTYPEID` and `ProductLength` columns, I identified 128 cases where the `PRODUCTID` was duplicated (asides first occurrence)
- I kept the first occurrence, and deleted the duplicates, to ensure each Product ID is unique.
- Final Count: The dataset was reduced from 3,847 to 3,649 unique products.

c. Standardizing Data Formats

- *Column Names*: I standardized all headers to snake_case (e.g., `PRODUCTID` to `product_id`, `ProductLength` to `product_length`) for easier coding.
- *Data Types*:
 - I converted `product_id` and `product_type_id` to String to prevent it from being treated as a math variable.
 - I ensured `product_length` is stored as numbers rounded to 2 decimal places. Initially it had a mix of decimals and integers.

d. Verifying Data Accuracy

- Checked for negative lengths: None found.
- All `product_length` values are positive, though the extreme outliers (up to 96,000) still exist and may need capping depending on your specific use case.

4. Creating the `short_title` Feature

- I created a column `short_title` just besides title column
- Since I am dealing with 3000+ rows, manually trimming each title is not efficient, so I used this text formula:
`=LEFT(TRIM(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(B2, "Set of", ""), "includes", ""), "features", ""), "|", "-"), "PCS", "")), 47) &
IF(LEN(TRIM(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(B2, "Set of", ""), "includes", ""), "features", ""), "|", "-"), "PCS", "")))>47, "...", "")`
- How the formula works:
 - `SUBSTITUTE`: It acts like a series of "Find and Replace" commands. It hunts down promotional fluff like "Set of", "includes", and "features" and replaces them with nothing (""). It also changes clunky pipe symbols (|) into clean dashes (-).
 - `TRIM`: It removes any awkward double spaces or trailing spaces left behind by the removed words.

- `LEFT(..., 47)`: It forces the title to stop at exactly 47 characters.
- `IF(LEN(...) > 47, "...", "")`: If the cleaned title was longer than 47 characters, it adds an ellipsis (...) at the very end so the total length maxes out at 50 characters.

5. Documentation and Reporting

- Asides from this process walkthrough, I created a Technical report for this project containing
 - Introduction and objectives.
 - Detailed steps for data cleaning and title optimization.
 - Summary of improvements, with examples of original and short titles.